Assignment 4

Hyperparameter Tuning for English to Hindi Translation (Transformer)

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1. Baseline Metrics

Before applying any hyperparameter tuning, the model was trained for 100 epochs using fixed settings. The results obtained are given below.

Metric	Value
Total Training Time	6800 sec
Final Training Loss	0.0991
BLEU Score	0.7698

2. Hyperparameters Chosen for Tuning

The following four hyperparameters were selected for tuning using Ray Tune with the AsyncHyperBand scheduler. These were chosen because they have the most direct impact on model performance and training stability.

Hyperparameter	Range / Options	Reason for Choosing
Learning Rate	1e-5 to 1e-3	Most critical factor affecting convergence speed and stability
Batch Size	16, 64	Affects gradient quality and GPU memory usage
FFN Dimension (d_ff)	1024, 2048	Controls the capacity of feed-forward layers
Dropout	0.1 to 0.4	Helps control overfitting during training
Number of Attention Heads	4, 8	Affects how many patterns the attention mechanism learns

3. Best Configuration Found

After running the tuning trials, the following configuration gave the lowest validation loss.

Hyperparameter	Best Value
Learning Rate	1.17e-04
Batch Size	16
FFN Dimension (d_ff)	1024
Dropout	0.3774
Number of Attention Heads	8

4. Final Metrics of Best Model

The best configuration was then used to train the model fully. The results are compared with the baseline below.

Metric	Baseline (100 epochs)	Best Model
Training Time	6800 sec	2789 sec
Final Loss	0.0991	0.3620
BLEU Score	0.7698	0.89
Epochs to Match Baseline BLEU	100	50

5. Observations

The following observations were noted during the tuning process:

- Among all the hyperparameters tuned, the learning rate had the most noticeable effect on training. Trials with learning rate above 1e-3 showed unstable loss in the early epochs, while values around 1.17e-04 gave smooth and consistent convergence.
- Trials using d_ff = 2048 consistently achieved lower final loss compared to d_ff = 1024, but took roughly longer per epoch. The best configuration used d_ff = 1024, which shows that model capacity matters more than speed for this translation task.
- The baseline model took 100 epochs to reach a BLEU score of 0.7698. The best-tuned model matched this score in approximately 0.89 epochs, saving around 45 minutes of training time. This confirms that proper hyperparameter selection reduces unnecessary training significantly.

Model link: Jahanvi16/en-to-hi-transformer

GitHub Link: https://github.com/jahanvi0106/MLOps-Jahanvi_Gajera-M25CSA012/tree/Assignment4